



The history of volcanism on Venus

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ABSTRACT

Completion of a global geological map of Venus has provided documentation of the relative age relationships, spatial distribution, and topographic configuration of the major geologic units and permitted us to address several important problems concerning the volcanic history of Venus. We use these data to: 1) assess the stratigraphic position of volcanic units and landforms, 2) determine their relationship with structure and tectonics, 3) identify changes in volcanic style, flux and activity with time, and 4) determine the topographic configuration and stratigraphic position of volcanism in relation to the evolution of long-wavelength topography.

Tectonic Associations: The scale and abundance of tectonic structures helps to divide the spectrum of volcanic units into two main groups: 1) volcanic units where tectonic structures played a subordinate role and 2) tectonized units/terrains, in which tectonic structures are the primary features relative to volcanism.

Sequence of Volcanism: Volcanic units embay the majority of the tectonized terrains and a sharp transition from heavily deformed units to mildly tectonized volcanic plains indicates that a tectonically driven regime dominated the earlier stages of the geologic history of Venus. This was followed by a regime of predominantly volcanic activity during the middle stages of observed geologic history. The latter stage is characterized by a volcano-tectonic resurfacing regime.

Volcanic Units: Regional plains are the most widespread volcanic unit and are likely to have an average thickness of the order of 400–500 m. Buried and partly buried ‘ghost’ craters, seen commonly on Mars and Mercury, are very rare, strongly suggesting that the previous cratering record was erased prior to formation of regional plains. The vast plains show a very small number of obviously flooded craters, which strongly suggests massive volcanic flooding over large provinces. In contrast to regional plains, the stratigraphically younger and much less widespread lobate plains embay ~50% of craters interacting with them. This suggests that emplacement of lobate plains was more in equilibrium with the growing population of impact craters.

Relations with topography: There are two major groups of topographic highs on Venus: 1) plateau-like, tessera-bearing regions, and 2) dome-shaped and rifted rises. Tessera is the oldest stratigraphic unit and its association with the plateau-like highlands suggests that they formed near the beginning of the observable history during the tectonically dominated regime. Regional plains preferentially occur within the lowland regions. The correlation of the older tectonized units and the vast volcanic plains with this regional topographic pattern suggests that the major features of the long-wavelength topography of Venus (the plateau-like highs and the lowlands) formed prior to emplacement of regional plains. Lobate plains and rift zones postdate formation of regional plains and are closely associated with the dome-shaped rises. The characteristic features of lobate plains and rift zones match the gravity and topography signatures of the rises and suggest that they were active during the latest episodes of the history of Venus. The alignment of wrinkle ridges and the topographic configuration of the upper sub-unit of regional plains and lobate plains suggest that the beginning of formation of the rises somewhat overlapped the late stages of formation of regional plains.

Volcanic Styles: The main volcanic plains have different morphologies that indicate different volcanic styles. Small and abundant volcanic constructs of the older shield plains imply that their sources were pervasive and nearly globally distributed, but that the supply of magma at individual sources was limited. The steep-sided domes are spatially and stratigraphically associated with the shield plains. The small size of the constructs of shield plains and their association with the steep-sided domes are most consistent with shallow crustal melting and differentiation of magma in reservoirs and/or partial melting of the

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crustal materials. Regional plains that postdate shield plains are very abundant (~1/3 of the surface of Venus) and ubiquitous but their sources are not visible at the available resolution. These features strongly suggest that regional plains formed by voluminous volcanic eruptions from near globally distributed sources. This style of volcanic activity resembles that of terrestrial flood volcanism, but its scale is more easily reconciled with the decompression melting of a fertile mantle layer that underplated the crust. The style of volcanism of the younger lobate plains was distinctly different from those of shield plains and regional plains. The numerous flows of lobate plains suggest multiple episodes of volcanic activity. The areal distribution of lobate plains implies that their sources were discrete, that they formed in different areas at different times, and that some of them may represent feeders of radiating dike swarms. Several lines of evidence suggest that lobate plains formed during a prolonged time span from just after the emplacement of regional plains until geologically recent times.

Volumes and Fluxes: Estimates of the volume of the main volcanic units on Venus show that both shield plains and regional plains are the major contributors to the volcanic resurfacing on Venus. The total volume of volcanic materials erupted during the volcanically dominant regime is estimated to be from about 140 to 200 × 10⁶ km³. In sharp contrast to this, the total estimated volume of lobate plains is much smaller, ~20–30 × 10⁶ km³, corresponding to a volcanic flux that is about an order of magnitude smaller than the average intraplate volcanic flux on Earth.

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1. Introduction

Major interior processes, such as magmatism, are directly related to the loss of internal heat of the planets, which determines their thermal evolution and governs their geological histories to a large extent (Solomon and Head, 1982). Thus, unraveling of the history of volcanic activity is one of the keys for understanding the evolution of planetary bodies. Several specific first-order features of Venus indicate that this planet has great importance in contribution to our understanding of planetary volcanism.

- 1) Venus is almost as large as Earth and has both gravitational and compositional potentials for a long-lasting volcanism, in contrast to the smaller terrestrial planets (e.g., Moon, and Mars), where conduction dominates, lithospheres thicken relatively quickly, and volcanic activity is sparse in the last third of solar system history (e.g., Tanaka, 1986; Wilhelms, 1987). Comparison of the spatial and temporal patterns of volcanism on Venus and Earth can put important constraints on the geological evolution of the larger terrestrial planets, particularly in relation to early history.
- 2) Extrusive volcanic materials make up about 80% of the surface of Venus. Thus, unraveling the history of volcanism is crucially important for the understanding of geological evolution of Venus. There is a rich variety of volcanic landforms on Venus ranging from small (several kilometers in diameter) volcanic constructs to morphologically homogenous volcanic plains units thousands of kilometers in extent (e.g., Head et al., 1992; Crumpler and Aubele, 2000). Establishing typical associations and the stratigraphic position of these landforms are necessary steps toward the solution of the problem of the geodynamic evolution of Venus.
- 3) Atmospheric conditions on Venus (Moroz, 1983; Self, 1983; Avdukevsky et al., 1983) preclude the existence of liquid water near the surface and suppress wind activity. Both these factors strongly inhibit erosional/depositional processes (Arvidson et al., 1990, 1992). This feature of Venus is of an extreme importance because it narrows the spectrum of possible mechanisms that shape the morphology and topography of the surface and, in fact, it leaves room almost exclusively for the preservation and analysis of volcanic, tectonic, and impact processes.
- 4) The number of impact craters on the surface of Venus is small (Herrick et al., 1997; Schaber et al., 1998) and impact structures commonly do not obscure landforms of either volcanic or tectonic origin.

- 5) The very high atmospheric pressure on Venus inhibits gas exsolution, magma disruption, and pyroclastic volcanism (Head and Wilson, 1992), thus further preserving the record of extrusive volcanic activity.

Unraveling of the history of internal processes requires an understanding of the sequence of major events through the observable portion of the geological record. The small number of craters on Venus largely prevents the use of standard techniques for estimating unit absolute model ages and relative ages by crater counting (Neukum and Wise, 1976; Neukum et al., 2001). Thus, the assessment of relative ages of morphologically homogenous (at the scale of mapping) units based on stratigraphic embayment and crosscutting relationships plays a major role in the reconstruction of the sequence of events.

The stratigraphic approach was successfully applied for many individual areas of Venus (the USGS quadrangles of the program of the geological mapping of Venus) (e.g., Chapman, 1999; Bridges and McGill, 2002; Ivanov and Head, 2010). However, the isolated nature of the quadrangles and different mapping philosophies of the geologists who mapped specific regions resulted in a number of maps that often poorly match each other. Detailed analysis of the published geological maps (Ivanov and Head, 2011) shows two important features: (1) the majority of units mapped in different regions by different authors appear to be similar morphologically and hold the same stratigraphic position and (2) the general stratigraphic sequence documented in the published maps appears to be the same (Basilevsky and Head, 2000a). The sequence starts with tessera (Ivanov and Head, 1996; Guest and Stofan, 1999) and other tectonized units at the bottom of regional stratigraphic columns, followed by vast volcanic plains at the middle levels (Ivanov and Head, 2011), and more recently by a variety of volcanic flows at the top of the columns (e.g., McGill, 2000).

The recently compiled global geological map of Venus (Ivanov and Head, 2011) portrays the spatial distribution of specific volcanic and tectonic units. The global-scale correlation chart that summarizes the stratigraphic observations documented throughout entire surface of Venus accompanies this map (Fig. 1). This chart shows the same general stratigraphic relationships of units that have been established in many individual quadrangles (see the review and comparison in Ivanov and Head, 2011). In contrast to the mosaic of the quadrangles, however, the global map presents a coherent picture of units/structures and their stratigraphic relationships that characterize the entire planet. Thus, the

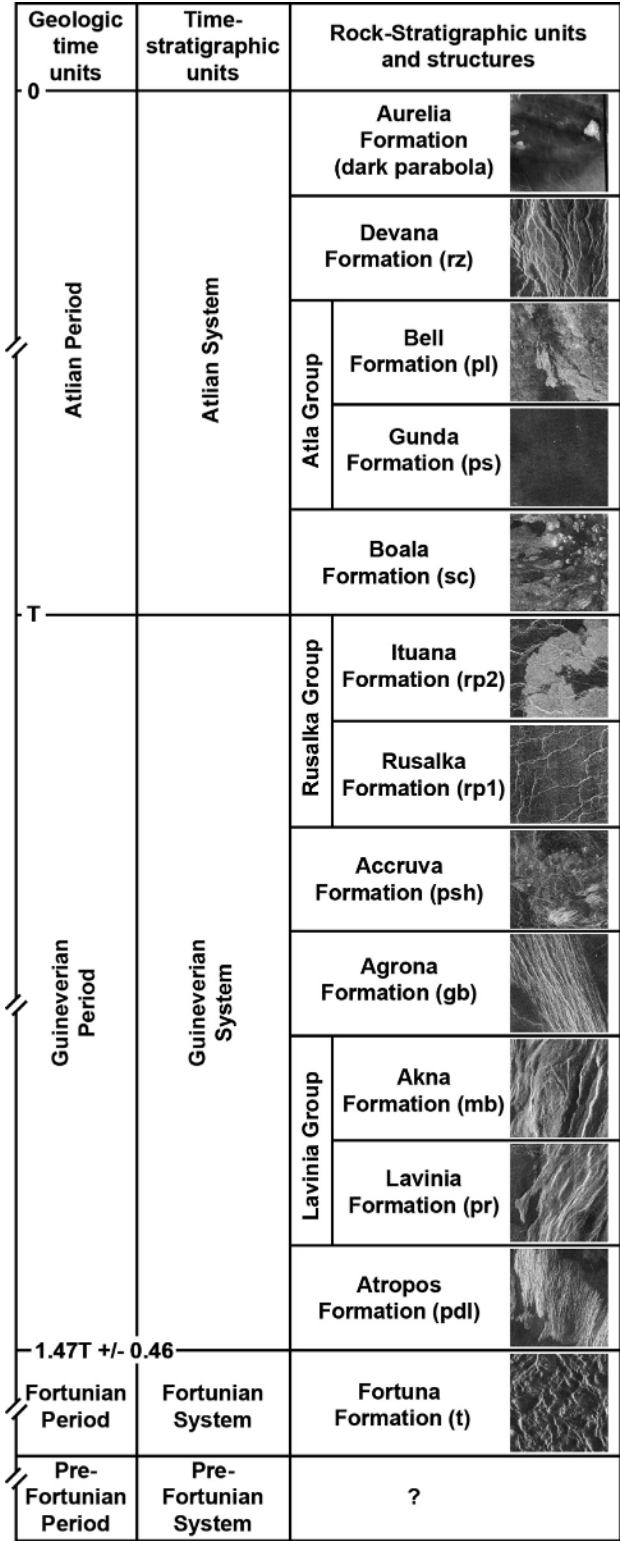


Fig. 1. The global stratigraphy of Venus based on the analysis of the relative age relationships among the units shown on the global geological map of Venus (reproduced from Ivanov and Head (2011)).

map provides a firm geological basis that allows assessment of the spatial and temporal distributions of landforms of different origin. Among related problems (for example, the role of mantle plumes, factors that control access of magma (Phillips and Hansen, 1994), the composition of volcanic units, etc.), the understanding of the history of volcanism on Venus includes several

important questions that are closely related to morphology, associations, and stratigraphy of volcanic features:

- 1) What is the variety of volcanic landforms on Venus, what is their petrogenetic significance, how do they contrast to the tectonized units/terrains, and what is their stratigraphic position?
- 2) What constraints do the relationships between the volcanic and tectonic units/terrains put on the nature of the observable part of the geologic history of Venus?
- 3) What style of volcanic activity do the volcanic landforms suggest and/or indicate? What does the stratigraphic position of volcanic landforms and their styles tell us about the evolutionary changes in volcanic styles and their relationship to the geodynamic evolution of the planet?
- 4) What constraints do the topographic configuration and stratigraphic position of the volcanic landforms introduce into the history of the long-wavelength topography?

The keys to these issues are the temporal, spatial, and topographic distribution of volcanic landforms that are summarized in the global geological map and correlation chart. In this paper we thus use the map as the basis to address the above problems and reconstruct the observable portion of the volcanologic history of Venus. We first describe the sequence of volcanic units, assess the nature of specific landforms that make them up, examine the age relationships, document the link to tectonic structures and features, and then examine their relationships to current and ancient topography. We conclude with a synthesis of the volcanic history of Venus, list insights that this provides and identify major outstanding questions.

2. Volcanic landforms on Venus and their typical associations

The specific morphology, dimensions of the occurrences, and relative importance of volcanic and tectonic processes permit the subdivision of the wide spectrum of landforms of Venus (Head et al., 1992; Crumpler and Aubele, 2000) into four broad categories: (1) vast and/or globally distributed volcanic units (plains), (2) local to regional volcanic features, (3) features that have both the volcanic and tectonic components (volcano-tectonic features), and (4) tectonized units/terrains, in which tectonic structures dominate the morphology. Several types of plains are globally important and the volcanic and volcano-tectonic features are often associated with these plains.

2.1. Volcanic plains

Volcanic plains are the most abundant volcanic features on Venus and occur throughout the global stratigraphic column of the planet (Fig. 1). The abundance of the secondary tectonic structures on the surface of the plains and the degree of their deformation define the following types of volcanic plains (Fig. 2).

2.1.1. Heavily tectonized plains (Fig. 2a and b)

Two units on Venus with the general aspects of volcanic plains have been so strongly deformed that tectonic structures clearly dominate and define the final morphology of the units. **Densely lineated plains (pdl, Atropos Formation, Fig. 2a; type localities are at: 67.2°N, 309.0°E; 43.8°N, 298.9°E):** numerous densely packed, narrow (from a few hundred meters wide and down to the resolution limit), short (a few tens of kilometers long), and parallel (subparallel) to each other lineaments heavily dissect the surface of this unit (Fig. 2a). If the lineaments are wide enough they usually appear as fractures. As a rule, the lineaments of pdl are packed so densely that they completely erase the morphology of the precursor materials. In some occurrences of the plains,